

USA TST 2026/1

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Problem: Let n be a positive integer. Prove that one can paint the non-zero coefficients of the polynomial

$$f(x_1, x_2, \dots, x_n) = \prod_{k=0}^n (x_1 + x_2 + \dots + x_n - k)$$

with $2^n - 1$ colors such that the coefficients of each color have sum 0, and each color is used at least once.

Proof. For ease, define the polynomials

$$P(x) = x(x-1)(x-2)\dots(x-n)$$

and $Q(x_1, x_2, \dots, x_k) = P(x_1 + x_2 + \dots + x_k)$. Consider any subset $S \subset \{1, 2, \dots, n\}$. For every such S , we define the color C_S . Suppose $S = \{s_1, s_2, \dots, s_t\}$. Then for every term of the form $x_{s_1}^{e_1} x_{s_2}^{e_2} \dots x_{s_t}^{e_t}$ we color its coefficient with the color C_S . Here, of course, all the e_i 's are greater than 0. Now clearly $2^n - 1$ colors are used. We prove that this works.

Claim: The sum of coefficients of the terms colored with the color C_S is 0.

Proof. Let $S = \{x_1, x_2, \dots, x_t\}$. The main idea is that the desired coefficients are a subset of the coefficients in the polynomial

$$P(x_1 + x_2 + \dots + x_t)$$

We now use strong induction on $|S|$. For the base case, WLOG let $S = \{x_1\}$. Then the coefficients of x_1^e are just the coefficients in $P(x_1)$, which clearly add up to 0 since $P(1) = 0$. Now for the inductive step, assume it holds true for all $|S| \leq t-1$. Then WLOG let $S = \{x_1, x_2, \dots, x_t\}$. Clearly if we consider the coefficients of the polynomial

$$P(x_1 + x_2 + \dots + x_t)$$

then the coefficients we are considering form a subset of these. However the remaining coefficients form all possible coefficients of the terms where $|S| \leq t-1$ and hence the sum of the coefficients under consideration is just

$$P(t) - c_1 P(t-1) - c_2 P(t-2) \dots - c_{t-1} P(1) = 0$$

where all the c_i 's are positive integers. Hence done. □

In fact we have

$$c_i = \binom{t}{i}$$

but that doesn't matter. □